

# MBot Control Board

## Ports & Pinout



For DSM Receiver

Power using either screw terminals or barrel jack 7.5V-12V

|       |          |          |         |       |
|-------|----------|----------|---------|-------|
| PWM0A | UART0_RX | I2C0_SDA | UART_RX | PIN 0 |
| PWM0B | UART0_TX | I2C0_SCL | UART_TX | PIN 1 |

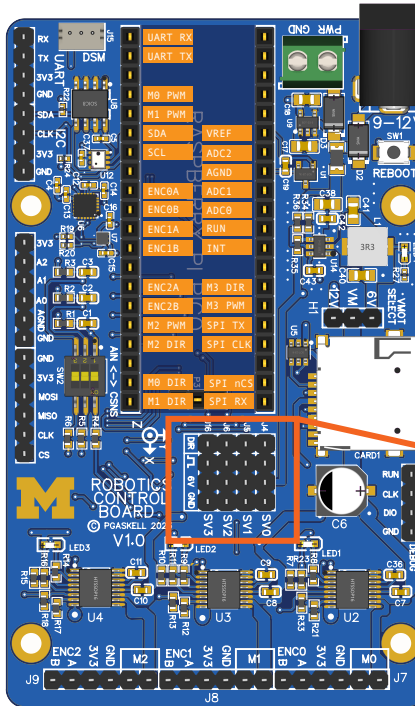
|       |          |          |         |       |
|-------|----------|----------|---------|-------|
| PWM2A | UART1_RX | I2C0_SDA | I2C_SDA | PIN 4 |
| PWM2B | UART1_TX | I2C0_SCL | I2C_SCL | PIN 5 |

I2C0 also connected to FRAM, IMU & Barometer

|       |          |      |      |        |
|-------|----------|------|------|--------|
| PWM6A |          | ADC2 | ADC2 | PIN 28 |
| PWM5B | I2C1_SCL | ADC1 | ADC1 | PIN 27 |
| PWM5A | I2C1_SDA | ADC0 | ADC0 | PIN 26 |

|       |          |          |          |         |        |
|-------|----------|----------|----------|---------|--------|
| PWM1B |          | I2C1_SCL | SPI0_TX  | SPI_TX  | PIN 19 |
| PWM0A | UART0_TX | I2C0_SDA | SPI0_RX  | SPI_RX  | PIN 16 |
| PWM1A |          | I2C1_SDA | SPI0_CLK | SPI_CLK | PIN 18 |
| PWM0B | UART0_RX | I2C0_SCL | SPI0_nCS | SPI_nCS | PIN 17 |

Select between current sense for M0-M2 and external Analog input.  
The current sense gives 1V/A.  
The ADC reads fullscale at 3.0V



Reboot the Pico without unplugging USB.  
To enter bootloader hold down both reboot and bootload and then release reboot.

Select Motor voltage 6.5V or VPOWER IN (12V)  
Servo power is always 6.5V

SD Card connected to SPI0  
nCS is multiplexed with M3 DIR

|       |          |          |        |        |        |        |          |          |       |
|-------|----------|----------|--------|--------|--------|--------|----------|----------|-------|
| PWM7A | I2C1_SDA | SPI1_CLK | M0_DIR | PIN 14 | PIN 2  | M0_PWM | I2C1_SDA | SPI1_CLK | PWM1A |
| PWM7B | I2C1_SCL | SPI1_TX  | M1_DIR | PIN 15 | PIN 3  | M1_PWM | I2C1_SCL | SPI1_TX  | PWM1B |
| PWM6B | I2C0_SCL | SPI1_nCS | M2_DIR | PIN 13 | PIN 12 | M2_PWM | I2C0_SDA | SPI1_RX  | PWM6A |
| PWM2B | I2C0_SCL |          | M3_DIR | PIN 21 | PIN 20 | M3_PWM | I2C0_SDA |          | PWM2A |

|          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| PIN 11   | PIN 10   | PIN 9    | PIN 8    | PIN 7    | PIN 6    |
| ENC2A    | ENC2B    | ENC1A    | ENC1B    | ENC0A    | ENC0B    |
| SPI1_TX  | SPI1_CLK | SPI1_nCS | SPI1_RX  | I2C1_SCL | I2C1_SDA |
| I2C1_SCL | I2C1_SDA | I2C0_SCL | I2C0_SDA | PWM3B    | PWM3A    |
| PWM5B    | PWM5A    | PWM6B    | PWM6A    |          |          |

Motor/Encoder Ports match Pololu 25mm gear motors with encoders

or MBot Encoder boards for Pololu 20mm gear motors